

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Electrical Engineering and Allied

Semester : IV

Subject Code & Name: BTBS404 Analog and Digital Electronics

Max Marks: 60

Date: 27/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- 1 In a common emitter amplifier configuration, which of the following is the most significant factor in determining the voltage gain?
a. Transistor's base current b. Load resistance c. Transistor's emitter current d. Voltage at the emitter
L1
- 2 What is the primary function of a transistor when used as an amplifier?
a. To switch between different states b. To convert DC signals to AC signals c. To amplify weak input signals d. To generate high-frequency signals
L1
- 3 Which of the following is a characteristic of an ideal operational amplifier?
a. Infinite open-loop gain b. Zero input offset voltage c. Infinite input impedance d. All of the above
L1
- 4 In the inverting configuration of an operational amplifier, the output signal is:
a. In phase with the input signal b. Out of phase with the input signal c. Unchanged by the amplifier d. Proportional to the input voltage but with no phase shift
L1
- 5 What is the binary equivalent of the hexadecimal number A3?
a. 10100011 b. 10000011 c. 11010011 d. 10110011
L1
- 6 Which Boolean law is used in the expression $A + AB = A$?
a. 0, 1, 1, 1 b. 1, 1, 0, 0 c. 1, 0, 0, 1 d. 1, 0, 1, 1
L1
- 7 Which of the following is a characteristic of TTL logic gates?
A) Low input impedance B) High output impedance C) High input current D) Low power consumption
L1
- 8 Which of the following is the typical application of a ring counter?
A) Digital clock generation B) Frequency modulation C) Memory storage D) Sequence generation in circuits
L1

9	Which of the following is the Product of Sums (POS) form of a Boolean expression?		1
	A) The sum of minterms	B) The sum of terms involving both literals and their complements	C) The product of maxterms
		D) The product of variables	L1
10	In the Karnaugh map for 3 variables, how many possible groups of minterms can be formed?		1
	A) 4	B) 6	C) 8
		D) 12	L1
	What is the primary function of a BCD to 7-segment decoder?		1
	A) To convert a binary number to decimal	B) To convert a BCD number to a 7-segment display format	C) To add two BCD numbers
		D) To convert a decimal number to BCD	L1
12	What is the main function of a demultiplexer (DE multiplexer)?		1
	A) It converts a binary signal to a decimal output	B) It selects one of many outputs and forwards it to the input	C) It takes a single input and directs it to one of many outputs
		D) It combines multiple signals into a single output	L1
Q.6	Solve the following.		12
A)	Derive four h parameters h_{11} , h_{12} , h_{21} , h_{22} in terms of the BJT's physical parameters.	CO1	6
B)	What is the 3 dB bandwidth of an amplifier, and why is it an important parameter in amplifier design?	CO1	6
Q.5	Solve the following.		12
A)	Define and explain the following terms commonly used in Op-Amp applications	CO2	6
	i. Power Supply Rejection Ratio (PSRR)		
	ii. Common-Mode Rejection Ratio (CMRR)		
	iii. Slew Rate		
B)	Define Op amp inverting amplifier with neat circuit diagram explain its working and derive the expression for gain.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain all logic gates with their symbols and truth tables	CO3	6
B)	Perform following binary arithmetic operations	CO3	6
	i. Multiply 1001 by 1101		
	ii. Divide 1110101 by 1001		
C)	Which gates are called Universal gates? Derive NOT, AND and OR gates from NAND gate.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain J-K flip-flop with circuit diagram and truth table.	CO4	6
B)	Compare TTL with CMOS Logic families.	CO4	6
C)	Reduce the expression for $f(A, B, C, D) = \sum m(1,5,6,12,13,14) + d(2,4)$ using K-map.	CO4	6

Q. 6 Solve Any Two of the following.

- A) Explain full adder circuit with the truth table.
- B) What is a decoder? Explain 2:4 decoders with the help of truth table.
- C) Explain 4-bit up counter with the help of truth table.

*** End ***

CO5
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